

EXPERIMENT – 10

Title

To Make Indexes of a Table

Aim / Objective

To understand the concept of **Indexes** in Oracle SQL and create, use, view, and drop indexes to improve the performance of data retrieval operations.

Theory

An **Index** is a database object that improves the speed of data retrieval from a table. It works similarly to the index of a book, allowing the database to locate rows quickly without scanning the entire table.

Indexes are created on one or more columns that are frequently used in **WHERE**, **ORDER BY**, **GROUP BY**, and **JOIN** clauses.

Although indexes improve query performance, they require additional storage space and slightly reduce the performance of INSERT, UPDATE, and DELETE operations because the index must also be updated.

Advantages of Indexes

- Improves query performance.
 - Reduces data retrieval time.
 - Speeds up JOIN operations.
 - Improves sorting and grouping performance.
 - Enhances overall database efficiency.
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Types of Indexes

1. Single Column Index

Created on a single column.

2. Composite Index

Created on multiple columns.

3. Unique Index

Ensures that indexed values are unique.

4. Bitmap Index

Used mainly in data warehouses for columns with low cardinality.

Learning Objectives

After completing this experiment, students will be able to:

- Understand database indexes.
 - Create single-column indexes.
 - Create composite indexes.
 - Display indexes.
 - Delete indexes.
 - Understand the advantages and limitations of indexes.
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Software Requirements

- Oracle Database
 - Oracle SQL Developer / SQL*Plus
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Algorithm / Procedure

1. Open Oracle SQL Developer.
 2. Connect to the Oracle database.
 3. Create the Employee table.
 4. Insert sample records.
 5. Create indexes.
 6. Display created indexes.
 7. Execute SQL queries using indexed columns.
 8. Drop indexes.
 9. Stop.
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Program

Step 1 : Create Employee Table

```
CREATE TABLE Employee
(
EmpID NUMBER PRIMARY KEY,
EmpName VARCHAR2(30),
Department VARCHAR2(30),
Salary NUMBER(10,2)
);
```

Step 2 : Insert Sample Records

```
INSERT INTO Employee VALUES(101,'Rahul','IT',35000);

INSERT INTO Employee VALUES(102,'Priya','HR',42000);

INSERT INTO Employee VALUES(103,'Aman','Finance',39000);

INSERT INTO Employee VALUES(104,'Sneha','Marketing',45000);

COMMIT;
```

Program 1 : Create a Single Column Index

```
CREATE INDEX IDX_EMPNAME
ON Employee(EmpName);
```

Display Index

```
SELECT
INDEX_NAME,
TABLE_NAME
FROM USER_INDEXES
WHERE TABLE_NAME='EMPLOYEE';
```

Expected Output

```
INDEX_NAME  TABLE_NAME
IDX_EMPNAME EMPLOYEE
```

Program 2 : Create Composite Index

```
CREATE INDEX IDX_EMP_DEPT  
ON Employee  
(Department,Salary);
```

Verify

```
SELECT  
  
INDEX_NAME  
  
FROM USER_INDEXES  
  
WHERE TABLE_NAME='EMPLOYEE';
```

Expected Output

```
INDEX_NAME  
  
IDX_EMPNAME  
  
IDX_EMP_DEPT
```

Program 3 : Create Unique Index

```
CREATE UNIQUE INDEX IDX_EMAIL  
ON Employee(EmpID);
```

Note

Since **EmpID** is already a **PRIMARY KEY**, Oracle automatically creates a unique index. This example is for demonstration purposes.

Program 4 : Retrieve Data Using Indexed Column

```
SELECT *  
  
FROM Employee  
  
WHERE EmpName='Rahul';
```

Expected Output

EmpID EmpName Department Salary

101	Rahul	IT	35000
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Program 5 : View All Indexes

```
SELECT  
  
INDEX_NAME,  
  
TABLE_NAME,  
  
UNIQUENESS  
  
FROM USER_INDEXES;
```

Expected Output

INDEX_NAME	TABLE_NAME	UNIQUENESS
IDX_EMPNAME	EMPLOYEE	NONUNIQUE
IDX_EMP_DEPT	EMPLOYEE	NONUNIQUE
SYS_C001245	EMPLOYEE	UNIQUE

Program 6 : Drop an Index

```
DROP INDEX IDX_EMP_DEPT;
```

Verify

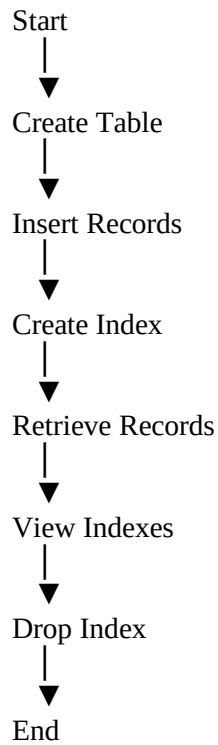
```
SELECT  
  
INDEX_NAME  
  
FROM USER_INDEXES  
  
WHERE TABLE_NAME='EMPLOYEE';
```

Expected Output

INDEX_NAME

IDX_EMPNAME

Flowchart



Explanation

CREATE INDEX

Creates a non-unique index on one or more columns.

CREATE UNIQUE INDEX

Creates an index that allows only unique values.

USER_INDEXES

Oracle system view used to display index information.

DROP INDEX

Removes an index permanently from the database.

Composite Index

An index created using more than one column.

Observation

- Single-column index was created successfully.
 - Composite index was created successfully.
 - Indexed columns improved query searching.
 - USER_INDEXES displayed all created indexes.
 - Index was deleted successfully using DROP INDEX.
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Result

Indexes were successfully created, displayed, utilized, and deleted. The experiment demonstrated how indexes improve query performance and support efficient data retrieval in Oracle databases.

Applications

- Banking Systems
 - Payroll Management Systems
 - Student Information Systems
 - Hospital Management Systems
 - Inventory Management Systems
 - Railway Reservation Systems
 - E-Commerce Applications
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Advantages

- Faster data retrieval.
- Improved search performance.

- Efficient sorting.
 - Faster JOIN operations.
 - Better overall query optimization.
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Limitations

- Requires additional storage space.
 - Slows down INSERT, UPDATE, and DELETE operations.
 - Requires maintenance when table data changes.
 - Too many indexes can reduce database performance.
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Viva Questions

1. What is an Index?
 2. Why are indexes used?
 3. What is the difference between a Primary Key and an Index?
 4. What is a Composite Index?
 5. What is a Unique Index?
 6. Which SQL statement is used to create an index?
 7. How do you display all indexes in Oracle?
 8. What is the purpose of USER_INDEXES?
 9. Can multiple indexes be created on a table?
 10. What are the advantages and disadvantages of indexes?
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Lab Assignment

1. Create a **Student** table and create an index on the Student Name column.
2. Create a composite index using Course and Marks columns.
3. Display all indexes created on the Student table.
4. Drop an existing index.
5. Compare Primary Key and Index.

6. Explain the difference between Unique and Non-Unique indexes.
 7. Write SQL queries using indexed columns.
 8. Study the execution plan before and after creating an index.
 9. Explain the applications of indexes in large databases.
 10. Create indexes for an Employee Management System.
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Precautions

- Create indexes only on columns that are frequently searched or used in JOIN conditions.
 - Avoid creating unnecessary indexes, as they increase storage requirements and slow down DML operations.
 - Do not create duplicate indexes on the same column.
 - Verify the existence of an index before dropping it.
 - Use USER_INDEXES and USER_IND_COLUMNS to inspect index details.
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Learning Outcome

After completing this experiment, students will be able to:

- Understand the concept and purpose of indexes in Oracle databases.
- Create single-column, composite, and unique indexes.
- View and manage indexes using Oracle data dictionary views.
- Improve query performance using appropriate indexing techniques.
- Apply indexing concepts in real-world database applications to optimize data retrieval.